CAT2 Marking Scheme

Computerized Database | Unit Code 0612 451 08A | 30 marks

Question 1 (2 marks)
A NULL value is a special marker in a database that represents missing, unknown, or inapplicable data. It is not the same as zero or an empty string.

Question 2 (2 marks)

- Adding records: INSERT
- Retrieving records: SELECT
- Modifying records: UPDATE
- Removing records: DELETE

Question 3 (2 marks)
A many-to-one relationship is one where many records in one table relate to a single record in another table.
Example: Many students belong to one course (many students → one course).

Question 4 (2 marks)

- **Primary indexing:** An index built on the primary key (the field the file is ordered by). It is unique and one per table.
- **Secondary indexing:** An index built on a non-primary-key field to speed up searches on that field. It may contain duplicates, and a table can have several.

Question 5 (2 marks)

- **Entity integrity:** Every table must have a primary key, and no primary key value may be NULL or duplicated, so each row is uniquely identifiable.
- **Referential integrity:** A foreign key value must either match an existing primary key value in the parent table or be NULL, so there are no orphan records.

Question 6 (4 marks)

a) One-to-one (2 marks)
Example: One person has one passport (or one employee has one staff ID card).
Implementation: The primary key of one table is placed in the other table as a foreign key that is also UNIQUE (or the two tables share the same primary key), so each record matches at most one record in the other table.

b) One-to-many (2 marks)
Example: One department has many employees (or one teacher teaches many students).
Implementation: The primary key of the "one" table (Department) is placed as a foreign key in the "many" table (Employee).

Question 7 (2 marks)
A primary key must uniquely identify every record. A NULL means unknown, so it cannot be compared or guaranteed to be unique, and a record with a NULL key could not be reliably identified or referenced by foreign keys. This is enforced by entity integrity.

Question 8 (4 marks)
Referential integrity is a rule that ensures relationships between tables stay consistent: every foreign key value in a child table must match an existing primary key in the parent table (or be NULL).

Actions when a referenced parent record is deleted (any two):

1. **RESTRICT / NO ACTION:** The deletion is rejected while child records still reference the parent.
2. **CASCADE:** The related child records are automatically deleted as well.
3. **SET NULL:** The foreign key in the child records is set to NULL.
4. **SET DEFAULT:** The foreign key in the child records is set to a default value.

Question 9 (2 marks)
SQL stands for **Structured Query Language**.
Reason widely used: It is a standard language supported by almost all relational database systems, and it is simple and powerful for creating, querying and managing data.

Question 10 (2 marks)
A many-to-many relationship is implemented by creating a third table, called a junction

(linking/associative) table. It contains the primary keys of both tables as foreign keys, often combined as a composite primary key. This converts the relationship into two one-to-many relationships.
Example: Student and Course are linked through an Enrolment table (student_id, course_id).

Question 11 (2 marks)

```
INSERT INTO Students (student_id, first_name, last_name, course, gpa)
VALUES (101, 'Amina', 'Otieno', 'IT', 3.6);
```

Question 12 (2 marks)

```
SELECT first_name, course
FROM Students
WHERE gpa > 3.0;
```

Question 13 (2 marks)

```
UPDATE Students
SET course = 'Business'
WHERE student_id = 101;
```

END OF MARKING SCHEME